SOFTWARE ENGINEERING LAB 1 - Requirements Engineering & UML Use-Case Modelling

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Section: 4H

Case Study Title: EV Charging Station Reservation System

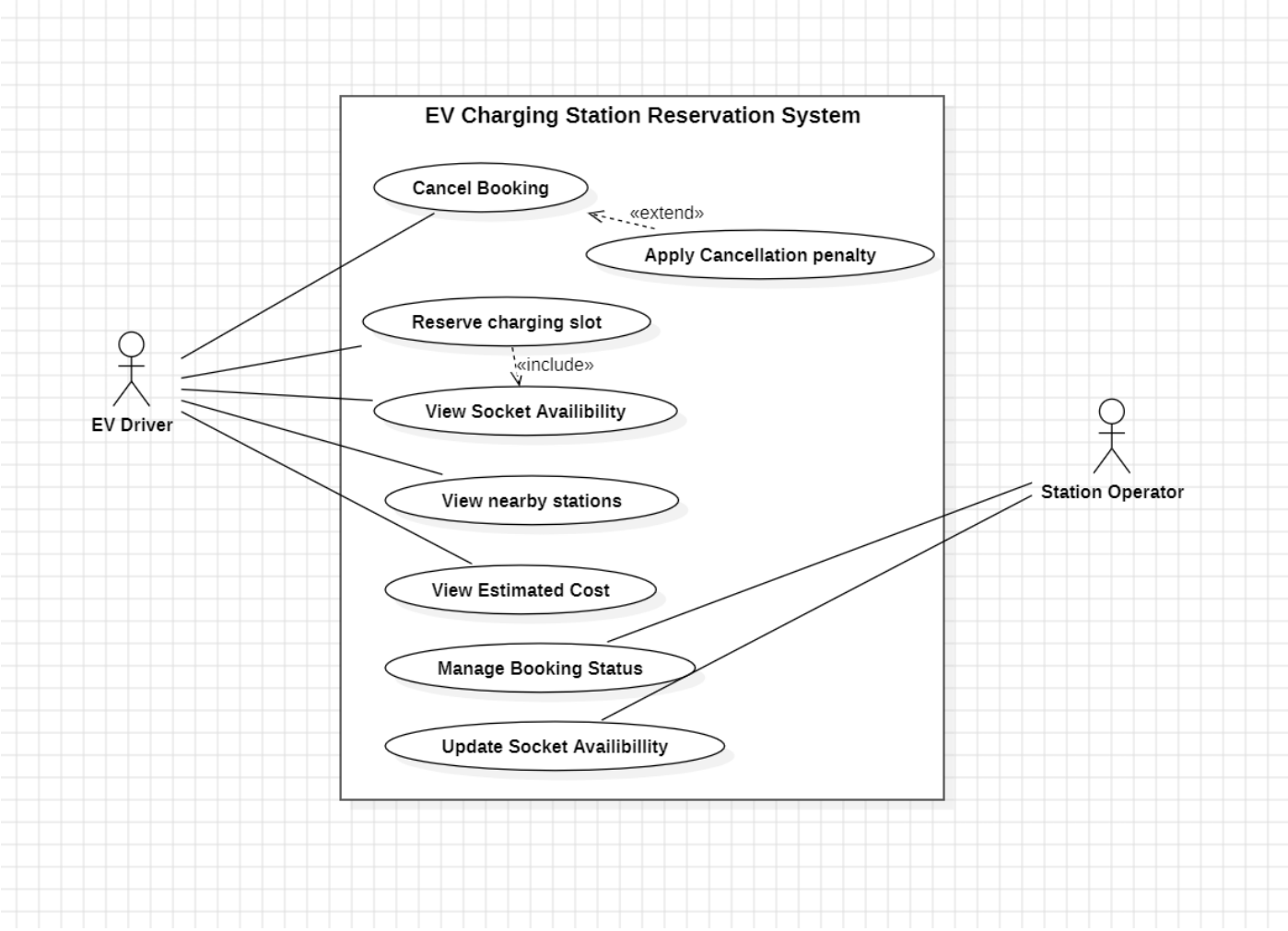
REQUIREMENTS TABLE:

Req ID	Type	Description	Priority	Acceptance	Rationale
FR-001	Functional	The system shall match the driver vehicle connector type with available station chargers and reserve a 30-minute charging window.	High	Pass: Incompatible connector plugs hidden from driver; Fail: Two vehicles booked for same socket concurrently.	Prevents drivers from booking or driving to a socket their car physically can't use, and stops two drivers from being double-booked on the same charger.
FR-002	Functional	The system shall display real-time availability (available/occupied/reserved) for each charging socket at a station.	High	Pass: Status on screen matches actual socket state within 5 seconds of a change; Fail: Screen shows available for a socket that is occupied.	Drivers need accurate info so they don't travel to a station that's actually full.
FR-003	Functional	The system shall calculate and display the estimated charging cost based on dynamic power rate before	Medium	Pass: Cost shown updates when rate changes and	Drivers should know the cost upfront so there are no

		the driver confirms a reservation.		matches the rate formula; Fail: Driver is charged a different amount than what was quoted.	billing surprises.
FR-004	Functional	The system shall allow a driver to cancel a reservation up to 5 minutes before the slot start time.	Medium	Pass: Cancelling within the window frees the socket immediately for others to book; Fail: A cancelled slot still shows as reserved/blocked.	Gives drivers flexibility and frees up chargers instead of wasting them.
FR-005	Functional	The system shall display all charging stations with compatible sockets within a 10 km radius of the driver's current location, sorted by distance.	High	Pass: Only stations within 10 km with a matching connector type appear on the map/list, ordered nearest-first; Fail: A station beyond 10 km is shown, or an out of range / incompatible station appears in the results.	Drivers need to find a reachable, compatible station quickly
NFR-001	Non Functional: Performance & Security	The system shall automatically release locked sockets and apply a penalty if the driver fails to plug in within 10 minutes of slot start.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load ; Fail: Slot stays locked/reserved past the 10-minute grace period with no penalty applied, blocking other drivers from using an available charger.	Stops drivers from hoarding a slot without using it, so chargers don't sit idle while others wait.
NFR-002	Non Functional:	The system shall support at least 500 concurrent	High	Pass: Load test with 500	Reservation systems get

	Performanc e	reservation requests without response time exceeding 3 seconds.		simultaneous users shows all responses under 3s; Fail: Any response exceeds 3s or requests fail under that load.	hit hard during peak commute hours, so it needs to hold up without lag or crashing.

UML USE-CASE DIAGRAM:



USE CASE FLOW SPECIFICATION:

Use Case: Reserve Charging Slot
System: EV Charging Station Reservation System
Primary Actor: EV Driver

Preconditions:

1. Driver is logged into the app.
2. Driver's vehicle connector type (CCS / CHAdeMO / Type 2) is registered on their profile.
3. Driver has a valid payment method on file.
4. At least one compatible socket is currently available .

Postconditions:

Success: A 30-minute charging window is reserved and locked for the driver's vehicle at the selected socket; the socket status changes to "Reserved"; a confirmation is displayed to the driver.

Failure: No slot is reserved; the socket remains available for other drivers; the failure reason is displayed to the driver.

Main Success Scenario:

1. Driver selects a nearby, compatible charging station from the results (View Nearby Stations).
2. System displays available sockets at that station («include»: View Socket Availability).
3. Driver selects an available, compatible socket and requests a reservation.
4. System checks that the driver's vehicle connector type matches the selected socket.
5. System calculates and displays the estimated cost for the reservation.
6. Driver confirms the reservation.

7. System locks the socket, creates a 30-minute reservation window, and updates the socket status to "Reserved."
8. System displays a confirmation to the driver, including station, socket, and time window.
9. Use case ends successfully.

Alternate Flow:

6a. Driver fails to plug in within 10 minutes of slot start («extend»: Apply Cancellation Penalty)

6a1. System detects that charging has not started within the 10-minute grace period.

6a2. System automatically releases the locked socket, making it available to other drivers.

6a3. System applies a no-show penalty to the driver's account.

6a4. Driver is notified that their reservation was cancelled and a penalty was applied.

6a5. Use case ends in failure.